

Tokenization

Manish Shrivastava

What are tokens?

- Tokens are "atomic" units of text which are processed by 🤖 ?
- Sort of,
- How many tokens in sentence below,

	1	2	3	4	5	6	7	8	9	10	11
1	In	the	winter	that	seat	is	close	enough	to	the	radiator
2	to	remain	warm,	and	yet	not	so	close	as	to	cause
3	perspiration.	In	the	summer	it's	directly	in	the	path	of	a
4	cross	breeze	created	by	open	windows	there,	and	there.	It	faces
5	the	television	at	an	angle	that	is	neither	direct,	thus	discouraging
6	conversation,	nor	so	far	wide	to	create	a	parallax	distortion,	I
7	could	go	on,	but	I	think	I've	made	my	point.	

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What are tokens?

- What about : it's or I've?
- What about : face and faces?

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Type-Tokens

- Type: face and faces => face
- Tokens: face and faces are two different tokens – for now.

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7	could	go	on	.	but	I	think	I've	made	my	point	.	

Challenges

- Hyphenation
 - Maybe, although the history of science-fiction is not on your side.
- Number
 - Let me do the math for you, this car weighs let's say 4,000lb, now add say 140 for me, 120 for you.
- Dates
 - June 30th, 2004. Opening day of Spider-man 2 at the AMC Pasadena. They only had red icees, no blue.
- Abbreviations
 - I'm Dr. Sheldon Cooper, B.S., M.S., M.A., Ph.D., and Sc.D. OMG, right?

Challenges

- Punctuations
 - I ... uh ... not sure how to proceed
- URLs
 - If she were to log on to www.socalphysicsgroup.org/activities/other, click on upcoming events,.....
- Sentences
 - Of course I do. My watch is linked to the atomic clock in Boulder, Colorado. It's accurate to one tenth of a second.
- Emojis
 -  (invisible joiners)
 - In reality a sequence of unicodes. As such can be a token, a naïve tokenizer might split the unicodes. Which behaviour we want? Depends on usage.

Challenges

- Code switching
 - Main market ja raha hoon, depending on tokenizer used, market might be only thing not split.
- Disfluency – Fragments
 - **You- uh-** y'know what, I'm gonna go. **I don't-** I don't think I can be involved in this particular thing right now.
- Disfluency – Fillers
 - How about this, um, okay, uh there's this farmer, and he has these chickens, but they won't lay any eggs.
- Tweets with extra emotions
 - I am happy!!!! 😄😄😄

So many words...

Corpus	Types = $ V $	Instances = N
Shakespeare	31 thousand	884 thousand
Brown corpus	38 thousand	1 million
Switchboard telephone conversations	20 thousand	2.4 million
COCA	2 million	440 million
Google n-grams	13 million	1 trillion

Why too many words is an issue?

- Many languages don't have orthographic words
- From POV NN how will you train a model vis-a-vis embedding layer?
- The number of words grows without bound (E.g. Parshendi, Alethi)

Split the Word__s

- Morpheme: Minimal unit with meaning in a language.
 - Face: one morpheme
 - Face__s: two morpheme *face* and *s*

Morphology: study of morphemes

Type of Morphemes

- Root: main meaning
- Affix (prefix/suffix): additional context and meaning
 - Work__ed
 - Glass_es

Types of morphemes

- Inflectional: added affix provides information like tense or number without changing the core meaning or class.
 - He *walked* till end of the road.
- Derivational: added affix changes the class of word.
 - Art of *googling*.

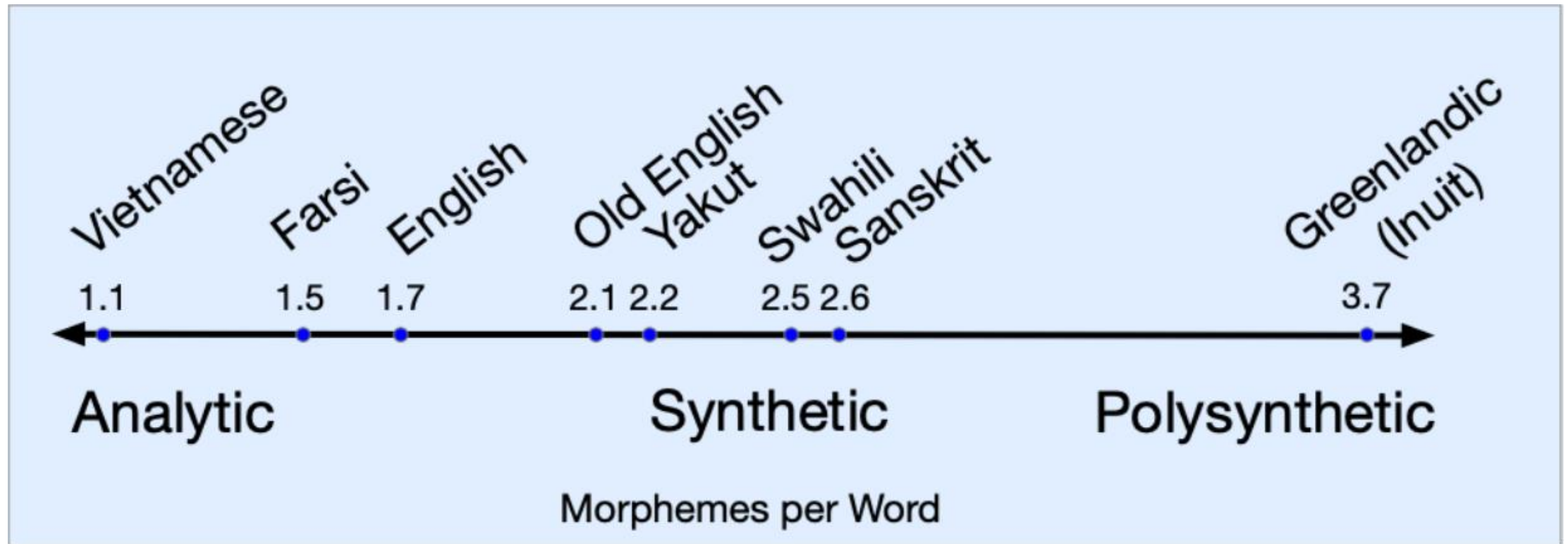
Clitics

- **Syntactically Independent:** Acts like a full word in a sentence.
- **Phonologically Dependent:** "Leans" on a host word for pronunciation.
- **Reduced Form:** A contracted or shortened version of a parent word.
 - I've or XYZ's book

Morphological Typology

- The study of how languages vary in their morphology, i.e., how words break up into their parts, is called morphological typology.
- Two important dimension for tokenization
 - Number of morphemes per word
 - How easily are morphemes separatable

Joseph Greenberg (1960) scale



How easily segmentable

- Agglutinative: like Telugu
 - ఇళ్ళకు (iḷḷaku)
 - ఇల్లు (illu) [House] + లు (lu) [Plural] + కు (ku) [To]
- Fusional:
 - English –s in "She reads the article"
 - Means both "third person" and "present tense"
- Morphemes are complex and require separate efforts for each language.

How to tokenize

- White space? Chinese, Japanese etc.?
- Morphemes? Language specific, hard to define
- Use Data to dictate the tokenization

Data for tokenization

- Morfessor
- Byte Pair Encoding
- Unigram Language Modeling tokenization

Morfessor

- Family of probabilistic machine learning methods for finding the morphological segmentation from raw text data.
- **Core Philosophy:** It mimics how a human might learn a language by identifying recurring meaningful units.
- Works by using Minimum Description Length (MDL), by balancing,
 - Keeping the list of stored morphemes as short as possible. Extreme cases
 - Consider a,b,c,d,... Very small vocabulary.
 - Consider cat, cats, walk, walking Very big vocabulary.
 - Representing the original text using as few morphemes as possible.
 - So for “Is the refrigerator working”
 - In first case text will have too many morphemes. I s t h e r e f r i g e r a t o r w o r k i n g
 - In second case text will have very few morphemes. Is the refrigerator working
 - Some where in between..

Baseline Algorithm

- Minimizing loss $L(S, D) = -\log P(S) - \log P(D|S)$
- Where,
 - $\log P(S)$: the prior probability of the lexicon. And penalizes large lexicons with many long, unique morphemes.
 - $\log P(D|S)$: is the probability of the corpus being generated by the morphemes in the lexicon. And penalizes overly granular segmentations.

Training

- Start by assuming every unique word in the data is a single "construction" (no splits).
- In each iteration, algorithm goes through every word in dataset.
- For a “token” (e.g. unfriendly) the algorithm tests **all possible two-part splits**:
 - u + nfriendly
 - un + friendly
 - unf + riendly ... and so on.
- If a split (like un + friendly) reduces the total cost, it accepts the split and immediately tries to split the resulting pieces recursively (e.g., checking if friendly should be friend + ly).
- Repeat until the total cost L stops decreasing.

Byte Pair Encoding

- Morfessor goes from word to character(s).
- BPE ([Sennrich et al. 2016](#)) comes from character(s) to word.
- It compresses data by merging frequent character pairs into symbols ([Gage 1994](#)).

Algorithm and Data Merge 0

Algorithm

For j in range (#merge operation)

- Get stat of each character bigram in vocab
- Get most frequent bigram, in case of multiple such bigram choose according to FCFS
- In vocab merge the most frequent bigram in a unigram

Data Status at Merge 0

- Vocab = {'l o w </w>': 5, 'l o w e r </w>': 2, 'n e w e s t </w>': 6, 'w i d e s t </w>': 3}
- Bigram count= {'(l', 'o)': 7, ('o', 'w)': 7, ('w', '</w>'): 5, ('w', 'e'): 8, ('e', 'r'): 2, ('r', '</w>'): 2, ('n', 'e'): 6, ('e', 'w'): 6, ('e', 's'): 9, ('s', 't'): 9, ('t', '</w>'): 9, ('w', 'i'): 3, ('i', 'd'): 3, ('d', 'e'): 3}
- Most frequent bigram = ('e','s'):9
- Vocab =
merged_bigram_with_vocab = {'l o w </w>': 5, 'l o w e r </w>': 2, 'n e w **e** s t </w>': 6, 'w i d **e** s t </w>': 3}

Algorithm and Data Merge 1

Algorithm

For j in range (#merge operation)

- Get stat of each character bigram in vocab
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- In vocab merge the most frequent bigram in a unigram

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- Bigram count= {('l', 'o'): 7, ('o', 'w'): 7, ('w', '</w>'): 5, ('w', 'e'): 2, ('e', 'r'): 2, ('r', '</w>'): 2, ('n', 'e'): 6, ('e', 'w'): 6, ('w', 'es'): 6, ('es', 't'): 9, ('t', '</w>'): 9, ('w', 'i'): 3, ('i', 'd'): 3, ('d', 'es'): 3}
- Most frequent bigram = ('es', 't'):9
- Vocab =
merged_bigram_with_vocab = {'l o w </w>': 5, 'l o w e r </w>': 2, 'n e w **est** </w>': 6, 'w i d **est** </w>': 3}

Algorithm and Data Merge 2

Algorithm

For j in range (#merge operation)

- Get stat of each character bigram in vocab
- Get most frequent bigram, in case of multiple such bigram choose according to FCFS
- In vocab merge the most frequent bigram in a unigram

Data Status at Merge 2

- Vocab = {'l o w </w>': 5, 'l o w e r </w>': 2, 'n e w e s t </w>': 6, 'w i d e s t </w>': 3}
- Bigram count= {('l', 'o'): 7, ('o', 'w'): 7, ('w', '</w>'): 5, ('w', 'e'): 2, ('e', 'r'): 2, ('r', '</w>'): 2, ('n', 'e'): 6, ('e', 'w'): 6, ('w', 'e s t </w>'): 6, ('w', 'i'): 3, ('i', 'd'): 3, ('d', 'e s t </w>'): 3}
- Most frequent bigram = ('l','o'):7
- Vocab =
merged_bigram_with_vocab = {'**l**o w </w>': 5, '**l**o w e r </w>': 2, 'n e w e s t </w>': 6, 'w i d e s t </w>': 3}

Algorithm and Data Merge 3

Algorithm

For j in range (#merge operation)

- Get stat of each character bigram in vocab
- Get most frequent bigram, in case of multiple such bigram choose according to FCFS
- In vocab merge the most frequent bigram in a unigram

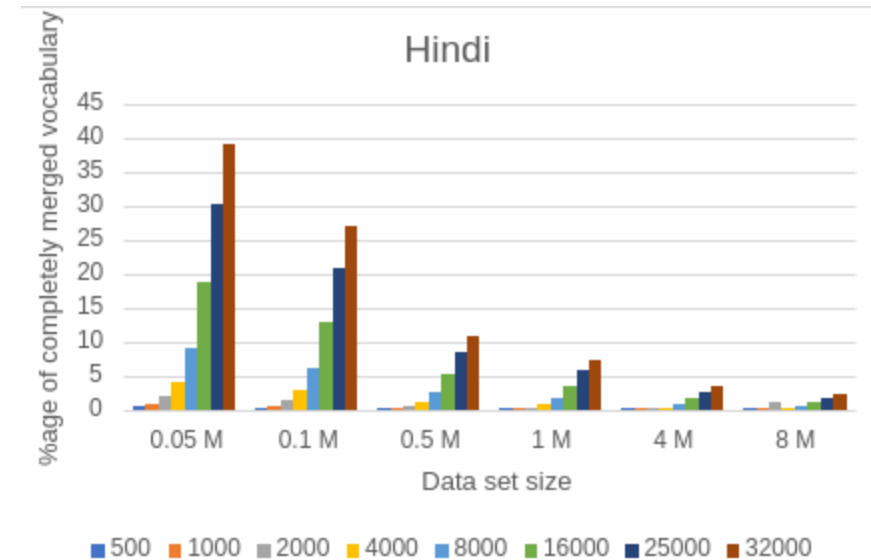
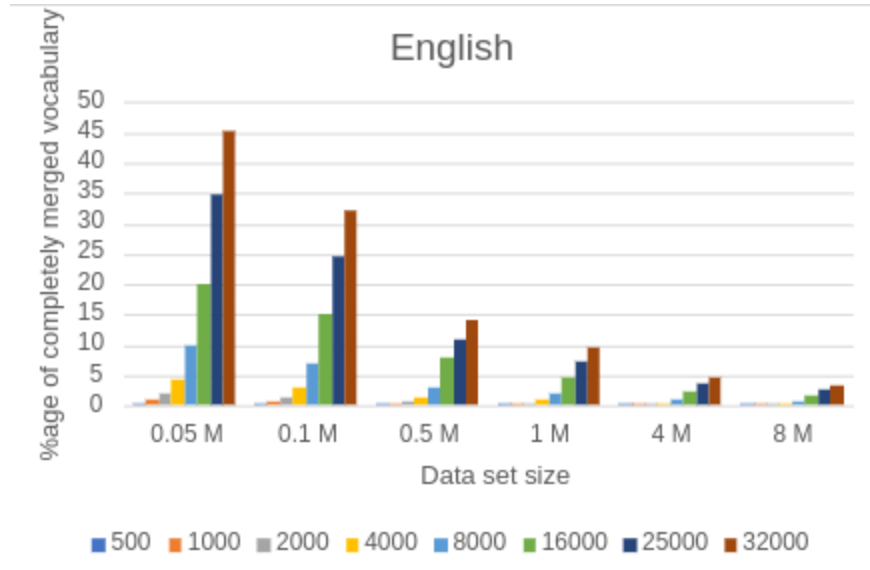
Data Status at Merge 3

- Vocab = {'lo w </w>': 5, 'lo w e r </w>': 2, 'n e w e s t </w>': 6, 'w i d e s t </w>': 3}
- Bigram count= {('lo', 'w'): 7, ('w', '</w>'): 5, ('w', 'e'): 2, ('e', 'r'): 2, ('r', '</w>'): 2, ('n', 'e'): 6, ('e', 'w'): 6, ('w', 'e s t </w>'): 6, ('w', 'i'): 3, ('i', 'd'): 3, ('d', 'e s t </w>'): 3}
- Most frequent bigram = ('l','o','w'):7
- Vocab =
merged_bigram_with_vocab = {'**low** </w>': 5, '**low** e r </w>': 2, 'n e w e s t </w>': 6, 'w i d e s t </w>': 3}

Tokenizing

- Once the model is trained, it has a **Ranked Merge List**. To segment a new, unseen word, the model applies these rules in the exact order they were learned during training.
- Now lets tokenize “lowest”:
 - Initial State: l_o_w_e_s_t
 - Step 1 (Apply l+o): lo_w_e_s_t
 - Step 2 (Apply lo+w): low_e_s_t
 - Step 3 (Apply e+s): low es_t
 - Step 4 (Apply es+t): low est

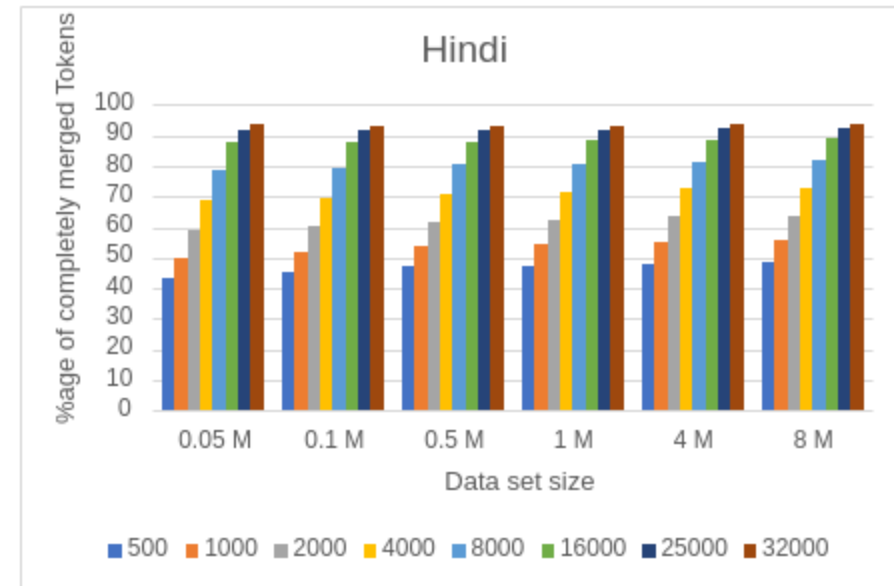
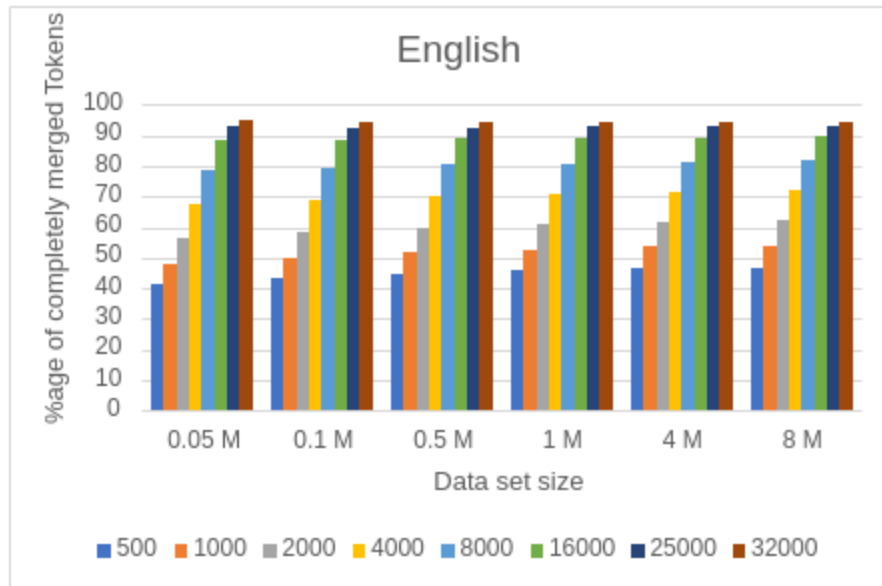
How many “words” in vocab are “complete”



Number of Merge Operations (NMO) 500, 1K, 2K, 32K

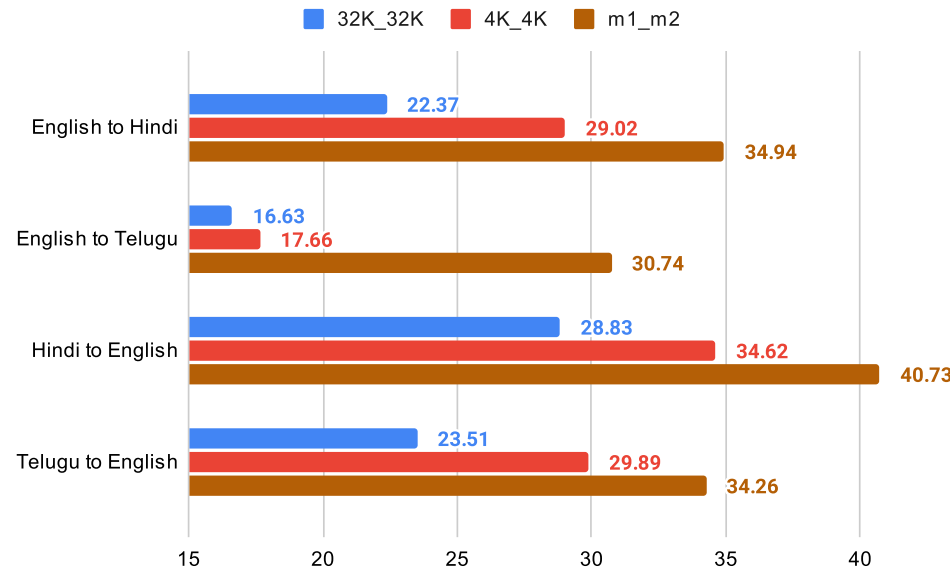
Dataset: [Samanantar](#)

How many “total words” are “complete”



Why should this granularity / imbalance matter?

- If lot of data is present sure go ham..
- But if data is less and we take traditional NMO (E.g. in MT 32K for each source and target language)?



On 0.1 Million, CHRF++ for evaluation

For more on how it effects in MT:
[Segmentation Beyond Defaults](#)

Unigram Language Modeling (sentencepiece)

- Models till now assume you can already split text into words using spaces.
- Doesn't work for languages like **Chinese, Japanese, or Thai**, where there are no spaces between words.
- ULM (sentencepiece) treats the **space** as just another character and uses `_` to represent it.

Sentencepiece

- Uses two different algorithms to decide how to "slice" the stream of characters.
 - These two characters appear together a lot; let's glue them into a token.
 - Iteratively **deletes** the least useful ones until it has efficient vocabulary.
 - $X = x_1, x_2, \dots$ Is subword sequence
 - x^* is most likely segmentation found using Viterbi algorithm.

$$P(\mathbf{x}) = \prod_{i=1}^M p(x_i),$$

$$\mathbf{x}^* = \arg \max_{\mathbf{x} \in \mathcal{S}(X)} P(\mathbf{x}),$$

$$\forall i \ x_i \in \mathcal{V}, \sum_{x \in \mathcal{V}} p(x) = 1,$$

- Thegoalkeeper
 - $P(\text{the}) = 0.08$ (Very common)
 - $P(\text{goal}) = 0.005$ (Common)
 - $P(\text{keeper}) = 0.002$ (Common)
 - $P(\text{go}) = 0.03$ (Very common)
 - $P(\text{alk}) = 0.0001$ (Rare fragment)
 - $P(\text{eeper}) = 0.0005$ (Suffix fragment)
- [The] + [goal] + [keeper]
 - $0.08 \times 0.005 \times 0.002 = 0.0000008$
- [The] + [go] + [alk] + [eeper](even though go is more likely than goal)
 - $0.08 \times 0.03 \times 0.0001 \times 0.0005 = 0.00000000012$

Why tokenization matters (MT example)

BPE configuration	32K_32K	4K_4K	16K_500	8K_500
Segmented Source	integrated machine translation systems are commonly used in social media services where the posts of users can be translated into the reader 's own language with a single click .	integr@@@ted mach@@@ine trans@@@lation systems are comm@@@only used in social media services where the po@@@sts of us@@@ers can be trans@@@lated into the read@@@er 's own language with a single cl@@@ick .	integrated machine translation systems are commonly used in social media services where the posts of users can be translated into the reader 's own language with a single click .	integrated machine transl@@@ation systems are comm@@@only used in social media services where the posts of users can be transl@@@ated into the read@@@er 's own language with a single click .
Output with Delimiters	एकीकृत मशीन प्रणाली में प्रयोग किया जाता है जहां सोशल मीडिया साइटों का प्रयोग अंग्रेजी में किया जाता है , जहां वे भाषा का प्रयोग कर सकते हैं ।	ए@@@ की@@@ कृत मशी@@@ न अनु@@@ वाद सोशल मीडिया में ही किया जाता है , जहां उप@@@ योग@@@ कर्ताओं के साथ एक ही भाषा के साथ अनु@@@ वाद किया जा सकता है ।	ए@@@ की@@@ कृत म@@@ श@@@ ी@@@ न अनु@@@ वा@@@ द सा@@@ मा@@@ न्य रूप से सो@@@ श@@@ ल मी@@@ डिया से@@@ वा@@@ ओं में प्र@@@ योग किया जाता है ज@@@ हां उप@@@ यो@@@ ग@@@ कर्@@@ ता@@@ ओं को एक ही क@@@ लि@@@ क के साथ पा@@@ ठ@@@ क में अनु@@@ वा@@@ द किया जा सकता है ।	ए@@@ की@@@ कृत म@@@ श@@@ ी@@@ न अनु@@@ वा@@@ द प्र@@@ ण@@@ ो@@@ लि@@@ यों का इ@@@ स@@@ ते@@@ मा@@@ ल सो@@@ श@@@ ल मी@@@ डिया से@@@ वा@@@ ओं में किया जाता है ज@@@ हां उप@@@ यो@@@ ग@@@ कर्@@@ ता@@@ ओं के प@@@ द@@@ ों को एक ही क@@@ लि@@@ क के साथ अनु@@@ वा@@@ द किया जा सकता है ।
Output with Glossary	एकीकृत मशीन प्रणाली → Integrated machine system में → in प्रयोग किया जाता है → is being used जहां सोशल मीडिया साइटों का प्रयोग अंग्रेजी में किया जाता है , → where social media sites are used in English जहां वे भाषा का प्रयोग कर सकते हैं । → where they can use the language	एकीकृत मशीन अनुवाद सोशल मीडिया में ही किया जाता है → Integrated machine translation is only used in social media जहां उपयोगकर्ताओं के साथ → where with users एक ही भाषा के साथ अनुवाद किया जा सकता है । → translation can be done in only one language	एकीकृत मशीन अनुवाद सामान्य रूप → Integrated machine translation commonly से सोशल मीडिया सेवाओं में प्रयोग किया जाता है → used in social media services जहां उपयोगकर्ताओं को एक ही क्लिक के साथ → where users, with one click पाठक में अनुवाद किया जा सकता है । → can translate into the reader	एकीकृत मशीन अनुवाद प्रणालियों का इस्तेमाल → Integrated machine translation systems are used सोशल मीडिया सेवाओं में किया जाता है → in social media services जहां उपयोगकर्ताओं के पदों को → where users' posts (here "posts" refers to positions in an organisation, not social media posts) एक ही क्लिक के साथ अनुवाद किया जा सकता है । → can be translated with a single click

Why tokenization matters (MT example)

Marathi Input	लशीचे दुष्परिणाम काय काय आहेत ?	English Input	You will see the following two options in it .
English Output	What side effects?	Marathi Output	खालील तुम्हाला दोन पर्याय दिसतील.
Marathi subword Input	लशी +चे दुष्परिणाम काय काय आहेत ?	English with subword	You will see the following two options in it .
English Output	What side effects of vaccines are there?	Marathi Output	तुम्ही दिसेल असे दोन पर्याय खालील दिली आहे.

Why understand how to tokenize

- One hyperparameter might not fit all.
- Varies depending on
 - languages,
 - dataset size,
 - tasks , etc.
 - Combinations of above.
- Tokens are “atomic” units of text which are processed by  ?
 - We define what atomic is, depending on task, model, language, budget etc.